

Finding and Filling Null Values in Independent Variables (X)

Find Null Values

The `isnull().sum()` method is used to count the number of missing (null) values in each selected independent variable. This helps identify incomplete data before model training.

```
#finding the sum of null values in the selected  
X.isnull().sum()
```

```
Country          0  
Life expectancy  5  
Mean years of schooling  7  
Gross national income (GNI) per capita  2  
Internet users    4  
dtype: int64
```

Code

```
X.isnull().sum()
```

Fill Null Values

Missing values are replaced with the mean of their respective columns using `fillna(X.mean())`. This ensures that all numerical features contain valid values without removing any records from the dataset.

```
#replacing the null values with the mean  
X.fillna(X.mean(),inplace=True)  
X.isnull().sum()
```

```
Country          0  
Life expectancy  0  
Mean years of schooling  0  
Gross national income (GNI) per capita  0  
Internet users    0  
dtype: int64
```

Code

```
X = X.fillna(X.mean())
```